

BPM CONSULTANT

THE FIRST STEP
IS TO IDENTIFY
YOUR PROBLEMS.



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WE DON'T HAVE
ANY PROBLEMS.
WHAT'S THE
SECOND STEP?



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MUST...
CONTROL
...FIST.

I HOPE
SOMEONE
GIVES ME
A BELT.



Announcements



NEWS

- Assignment #1 - late submissions accepted until 2/13 (50% credit)
- Quiz #1 on 2/11 - based on Lectures 3&4 (process modeling)

Modeling Example

Process Description (from “Discovery”)

A loan application is approved if it passes two checks: (i) the applicant's loan risk assessment, done automatically by a system, and (ii) the appraisal of the property for which the loan has been asked, carried out by a property appraiser. The risk assessment requires a credit history check on the applicant, which is performed by a financial officer. Once both the loan risk assessment and the property appraisal have been performed, a loan officer can assess the applicant's eligibility. If the applicant is not eligible, the application is rejected, otherwise the acceptance pack is prepared and sent to the applicant.

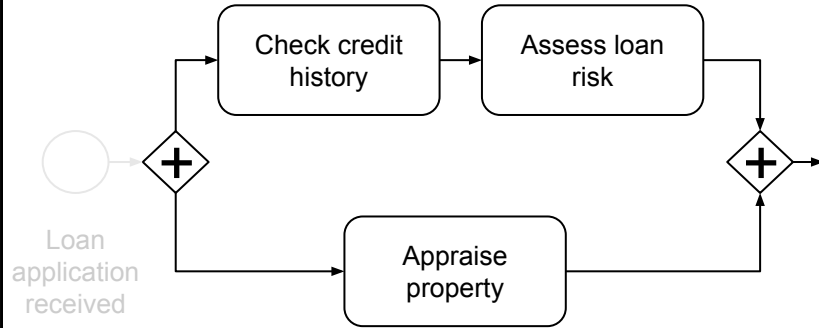
Step 1. Identify pool and lane

LOAN DEPARTMENT

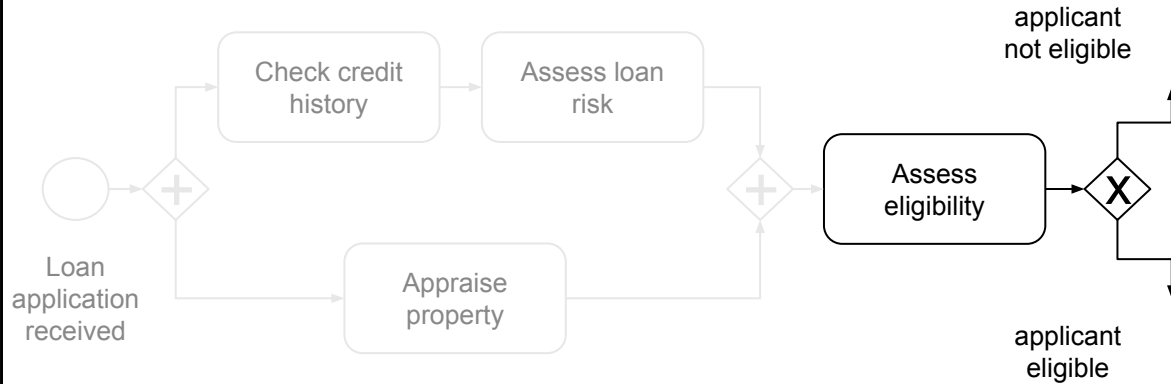
Step 2. Start process



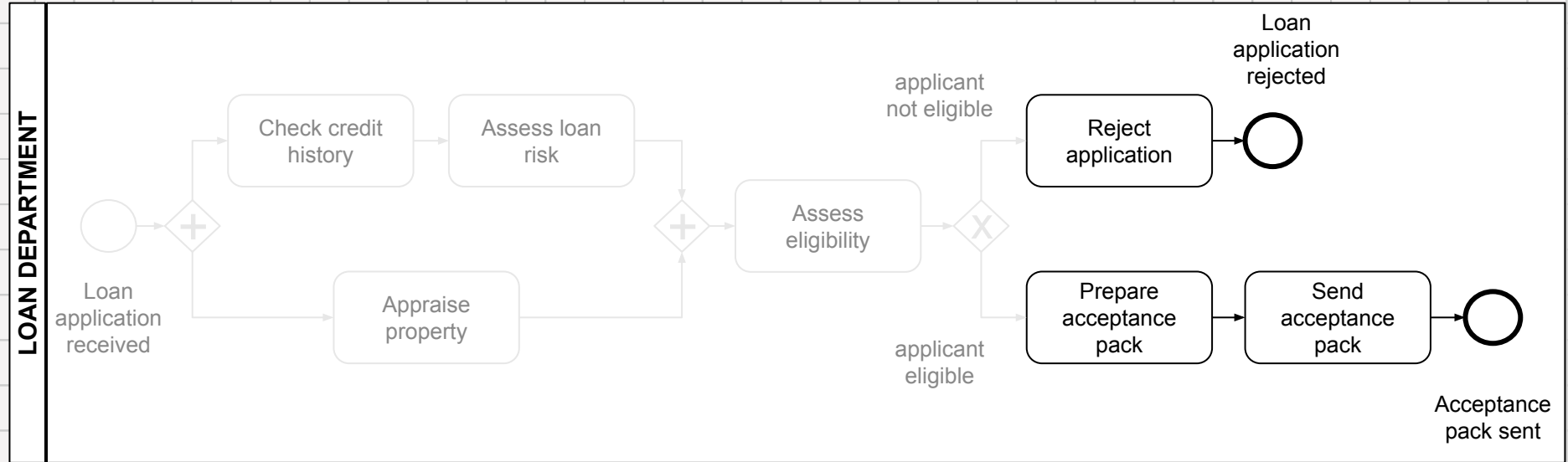
Step 3. two checks: i) credit history check followed by loan risk assessment **AND** ii) property appraisal



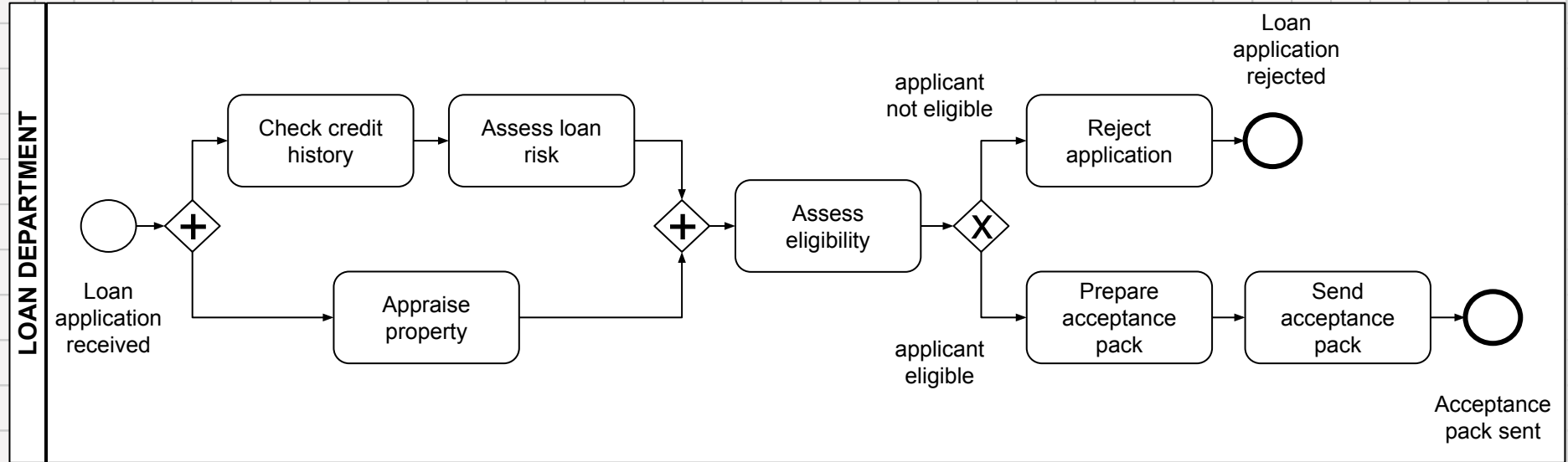
Step 4. Once both the loan risk assessment and the property appraisal have been performed, a loan officer can assess the applicant's eligibility.



Step 5. If the applicant is not eligible, the application is rejected, otherwise the acceptance pack is prepared and sent to the applicant.



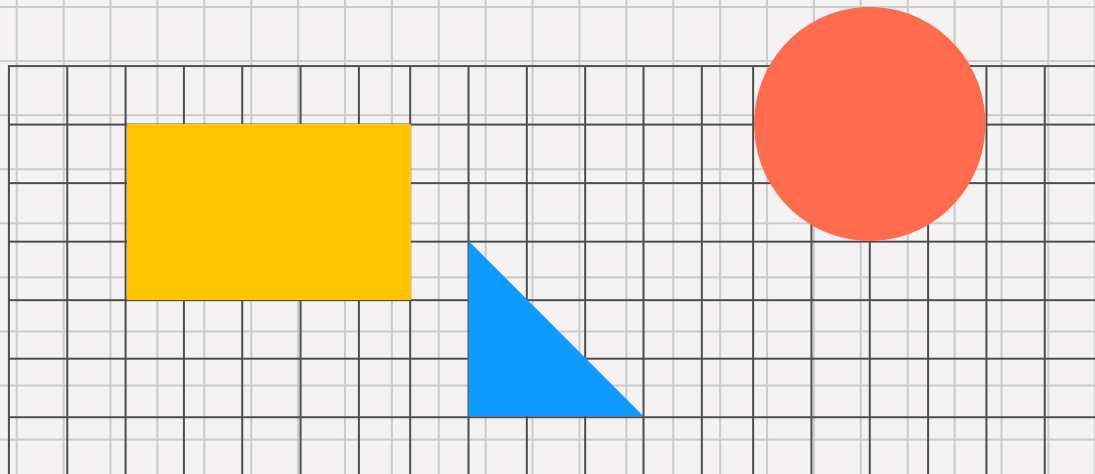
Step 6. Validate process map and finalize.



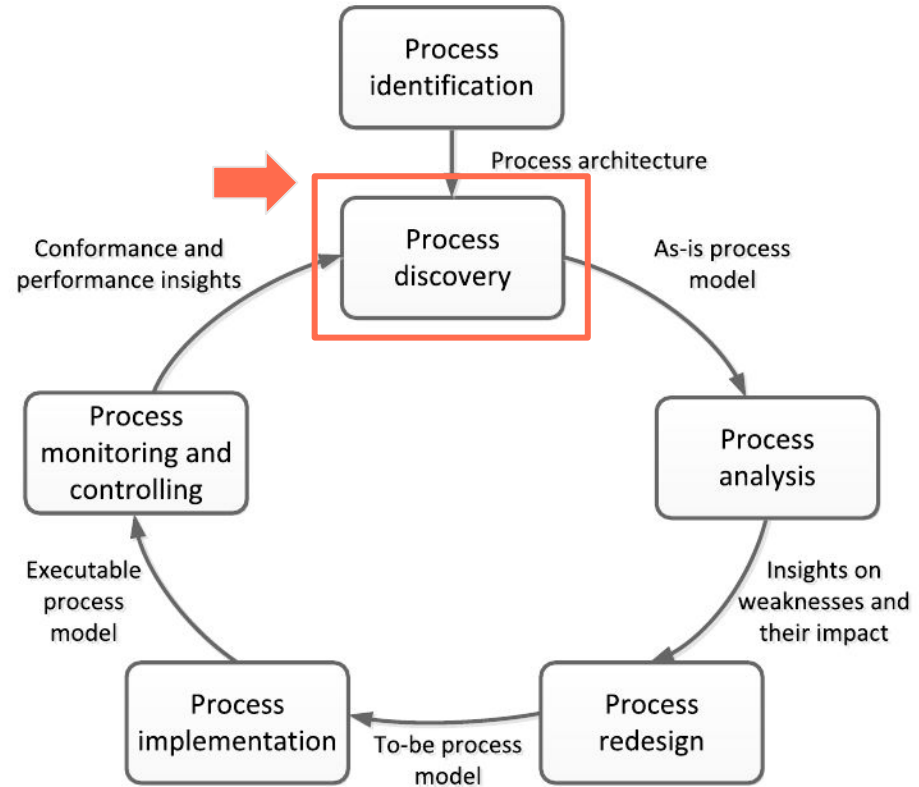
Spring 2026

IS 365

Lecture 5: Process Discovery



Focus Today





Introduction to Process Discovery

Process discovery is defined as the act of gathering information about an existing process and organizing it in terms of an as-is process model.

Why Process Discovery Matters

- Foundation for process analysis and improvement
- Captures 'as-is' reality before redesign
- Ensures alignment across stakeholders

Approaches to Discovery

- Evidence-based (documents, system logs, data)
- Interview-based (domain experts, stakeholders)
- Workshop-based (collaborative modeling sessions)

Aspect	Evidence-based	Interviews	Workshops
Objectivity	Which one is best?		
Richness			
Time consumption			
Immediacy of feedback			

Aspect	Evidence-based	Interviews	Workshops
Objectivity	High	Medium-high	Medium-high
Richness	Medium	High	High
Time consumption	Low-medium	Medium	Medium
Immediacy of feedback	Low	High	High

Challenges in Discovery

- Incomplete or inconsistent information
- Biases from participants
- Balancing detail vs. simplicity

Discovery Methods

Document Analysis

- Policies, manuals, system documentation
- Provides baseline process descriptions
- **Risk:** outdated or idealized information

Observation

- Shadowing workers in real settings
- Captures tacit knowledge and exceptions
- **Risk:** Hawthorne effect (behavior changes under observation)

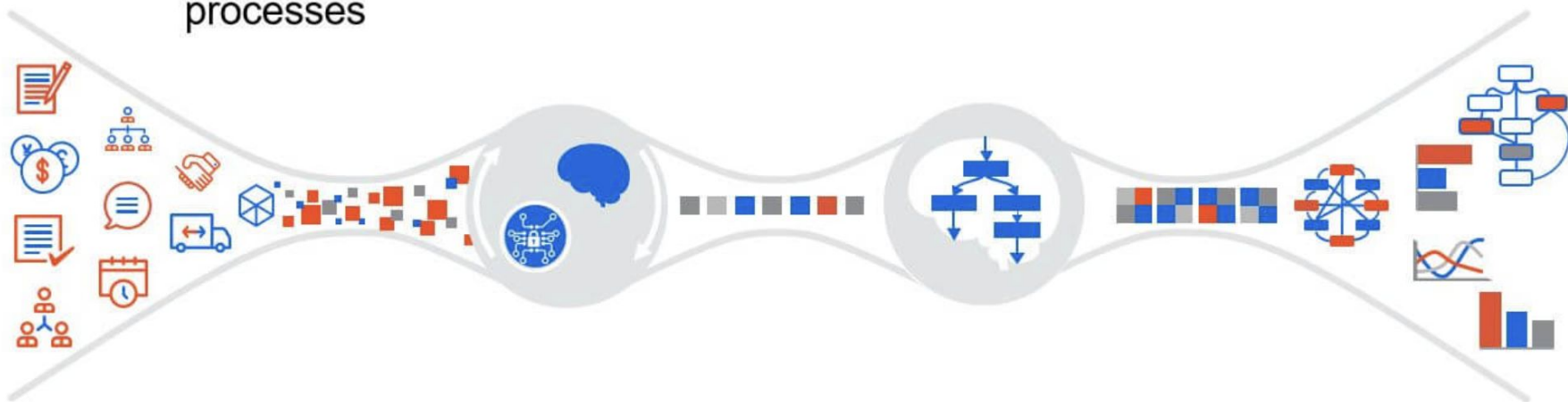
Process Mining / Automated Discovery

- Automated discovery from event logs
- Reveals actual execution paths and bottlenecks
- Complements human-driven methods

Reads
data related to your
processes

Transforms
data to event log

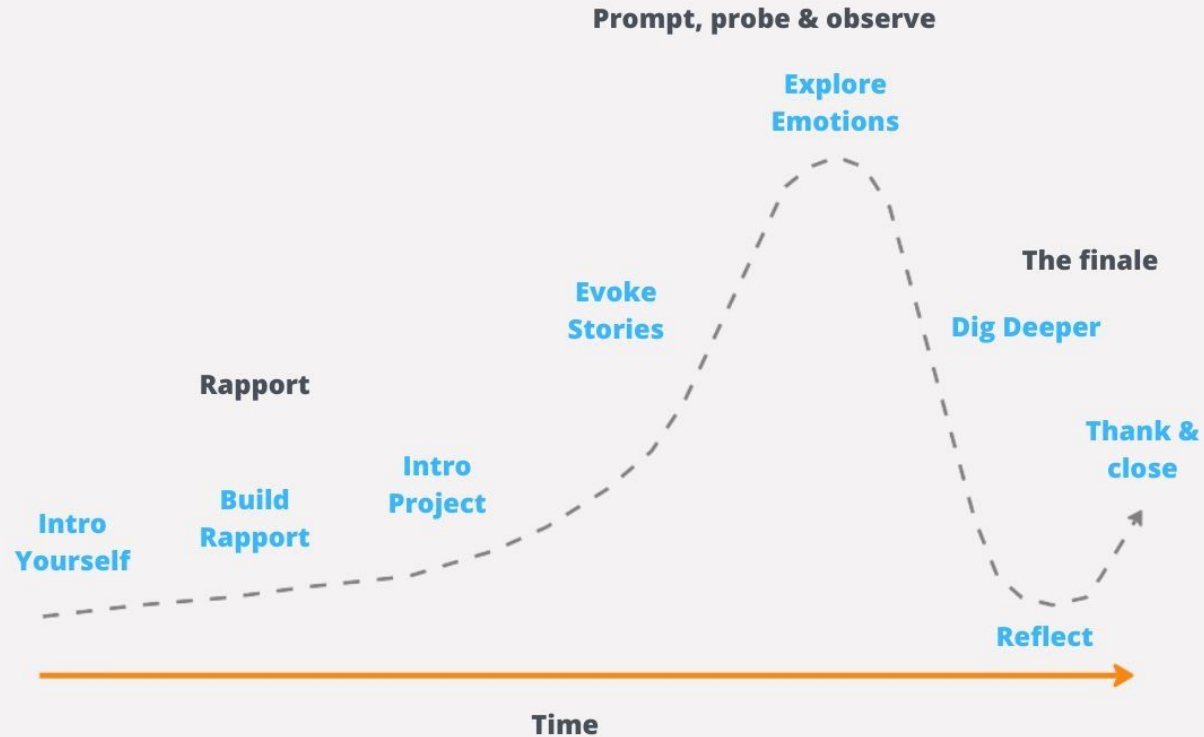
Visualizes
the end-to-end process



Interviews

- One-on-one with domain experts
- Flexible, detailed insights
- **Risk:** personal bias, limited scope

ANATOMY OF A DISCOVERY INTERVIEW



Workshops

- Group sessions with multiple stakeholders
- Builds consensus and shared understanding
- **Risk:** group think, dominance by stronger voices

Method	Strengths	Weaknesses
Document analysis	• Structured information • Independent from stakeholders availability	• Outdated material • Wrong level of abstraction
Observation	• Context-rich insights	• Potentially intrusive • Stakeholders likely to behave differently • Only few cases and not all processes can be observed
Automated discovery	• Extensive set of cases • Objective data	• Potential issue with data quality and level of abstraction • Data may not be available or be available only in part • Data extraction and preparation is time-consuming
Interviews	• Context-rich insights	• Requires sparse time of stakeholders • Time-consuming: several iterations required before sign-off
Workshops	• Context-rich insights • Direct resolution of conflicting views	• Requires simultaneous availability of multiple stakeholders • Time-consuming: multiple sessions typically required

Structuring Discovery

Stakeholder Identification

- Internal vs. external stakeholders
- RACI model (Responsible, Accountable, Consulted, Informed)
- Balancing perspectives

EXAMPLE OF A RACI CHART

	Project manager	Strategist	Designer	Front-end developer	Back-end developer
Create sitemap	C	R	A	I	I
Design wireframes	C	A	R	I	I
Create style guide	A	C	R	C	I
Code templates	A	I	C	R	C

R = Responsible

A = Accountable

C = Consulted

I = Informed

Scoping the Process

- Define clear boundaries (start & end points)
- Levels of granularity (high-level vs. detailed)
- Avoiding scope creep

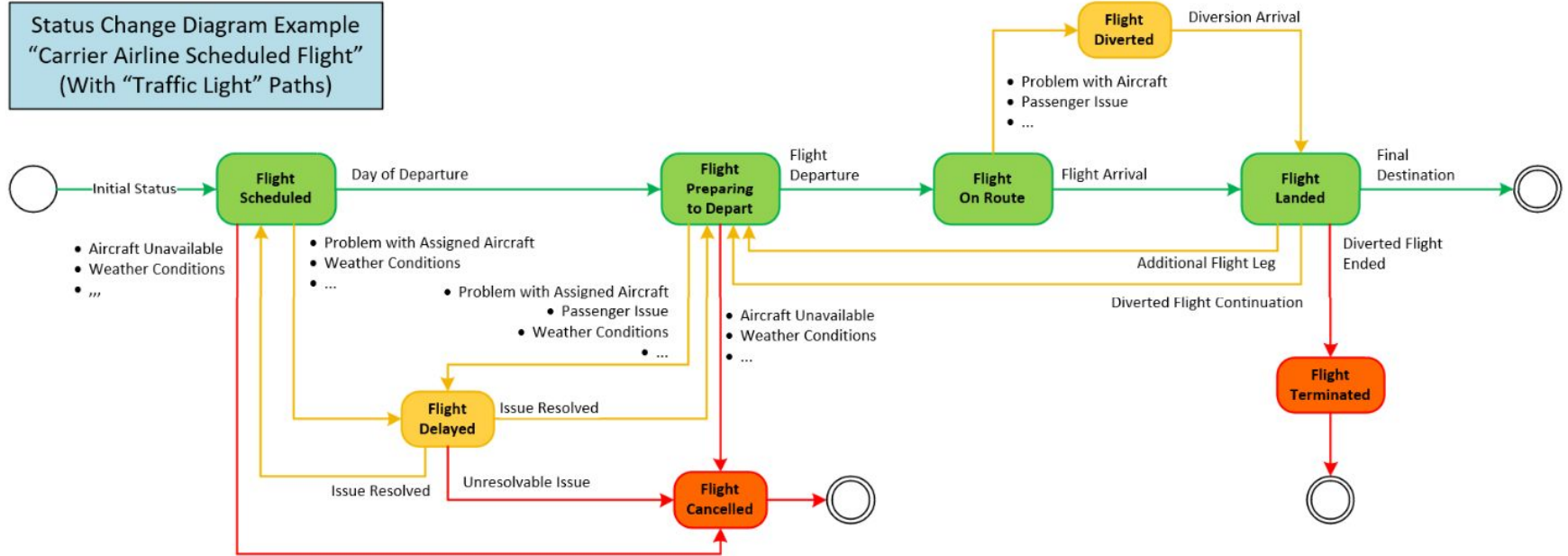
Facilitation Techniques

- Neutral facilitator role
- Encourage balanced participation
- Use visual aids (whiteboards, sticky notes, BPMN tools)

Capturing Variations

- Happy path vs. alternative paths
- Handling exceptions explicitly
- Identifying rework loops

Status Change Diagram Example "Carrier Airline Scheduled Flight" (With "Traffic Light" Paths)



From Discovery to Models

Translating Information to BPMN

- Use consistent symbols and naming conventions
- Validate models with stakeholders
- Capture both control flow and data flow


Purchase
order
received

Check stock
availability

Check raw
materials
availability

Retrieve
product from
warehouse

Request raw
materials

Obtain raw
materials

Manufacture
product

Get shipping
address

Ship
product

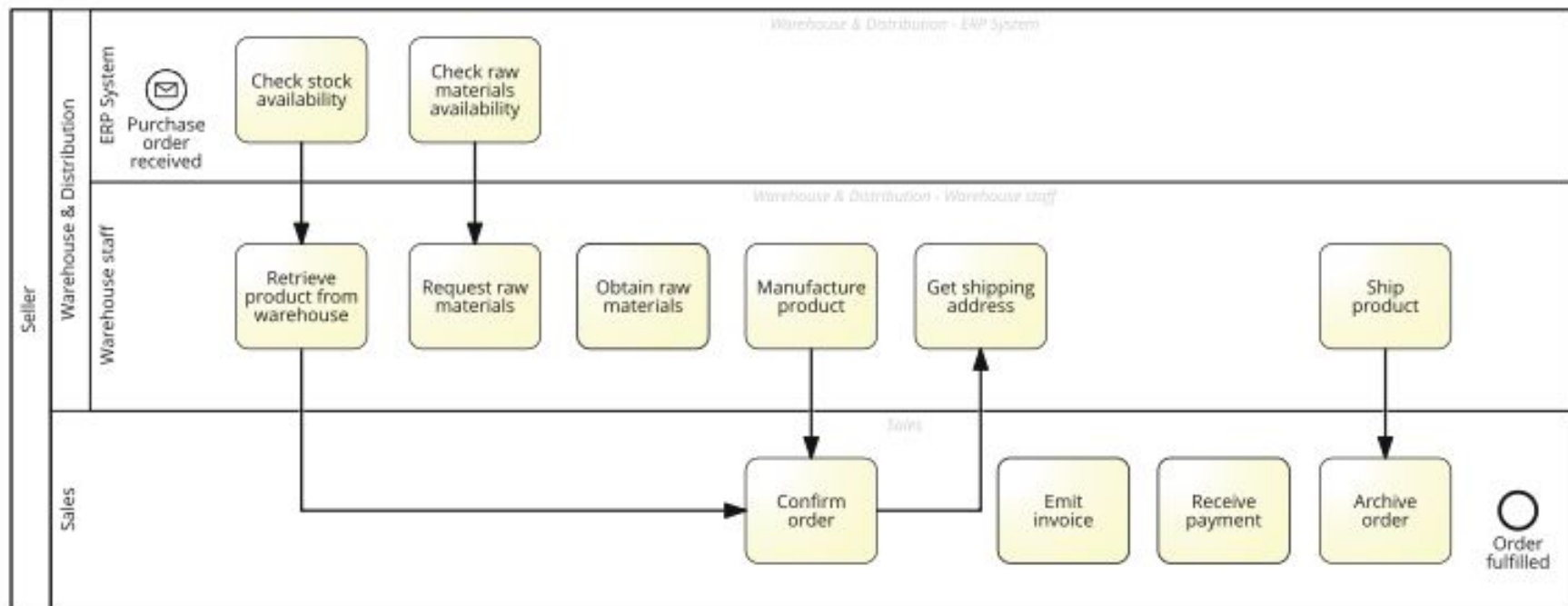
Confirm
order

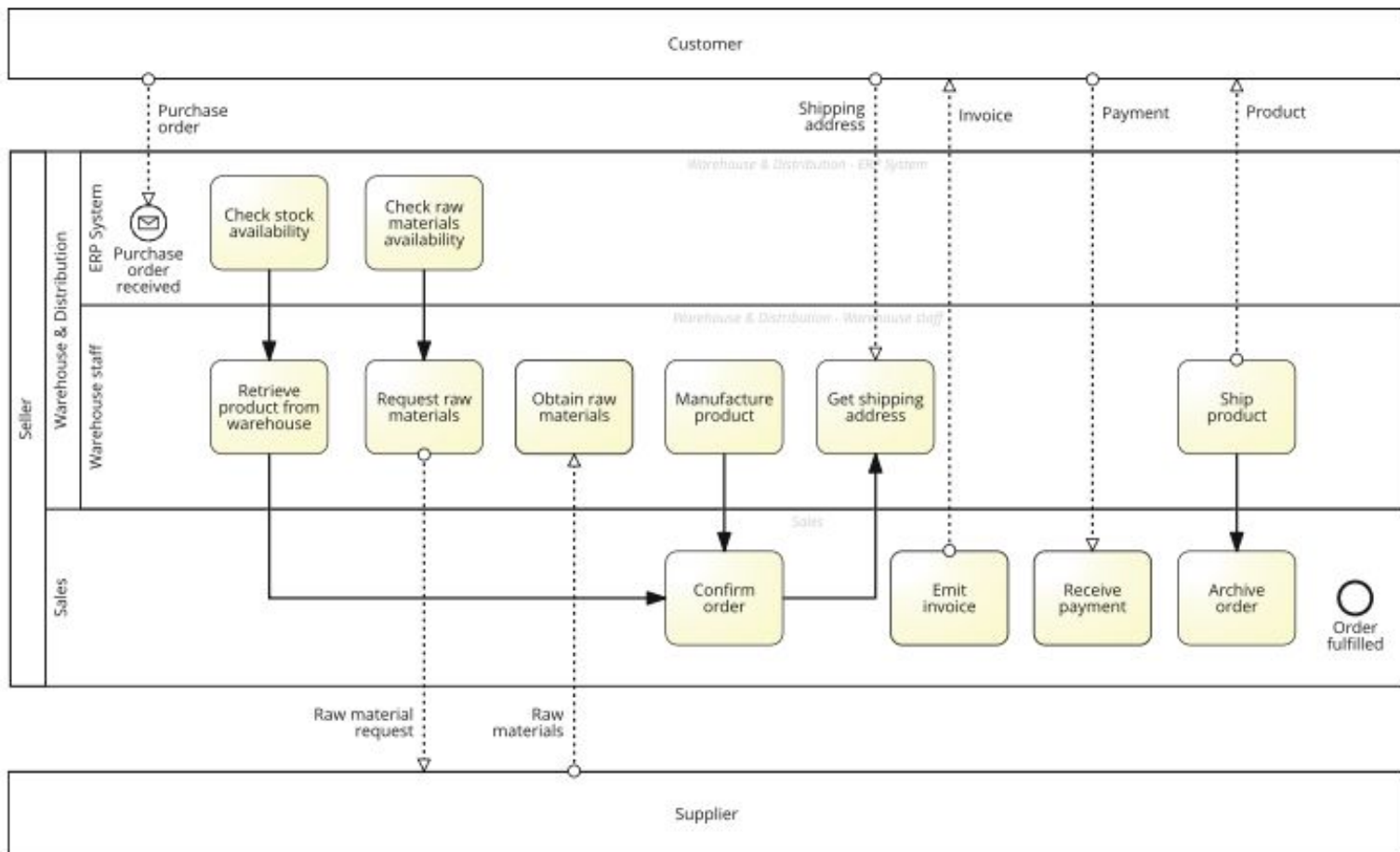
Emit
invoice

Receive
payment

Archive
order


Order
fulfilled





Validating the Model

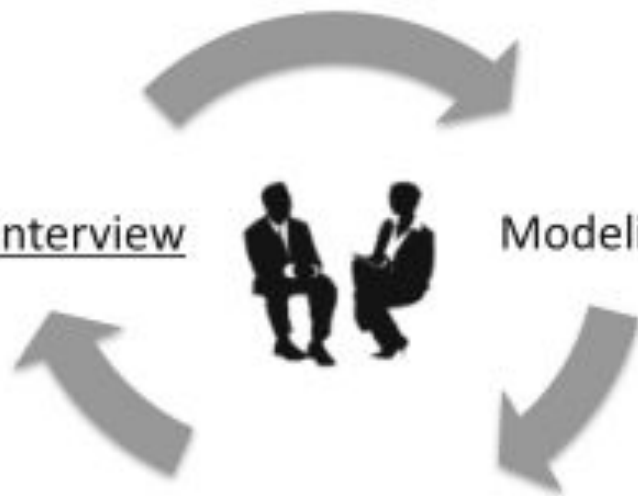
- Walkthroughs with stakeholders
- Role-play or simulation of scenarios
- Ensure completeness and correctness

Interview



Modeling

Validation



Common Pitfalls

- 
- Overcomplicating diagrams
 - Missing exceptions or handovers
 - Using jargon instead of business language

Best Practices for Discovery

- 
- Combine multiple methods (triangulation)
 - Iterative refinement of models
 - Continuous stakeholder involvement

Future of Process Discovery

Continuous Monitoring

- Continuous discovery using automated tools
- Extract understanding and implement actions in real-time
- Implement continuous improvement at the process level

Continuous Discovery

Actionability

Automation
Predictive analysis
Process optimization



Discovery

Process Discovery
Task Discovery
Automation Insights



Semantic Understanding

Semantic Extraction
Knowledge graph
Benchmarking

MONITORING

Continuously monitor process execution, provide KPIs in real time, and dynamically trigger bots to take recommended action.



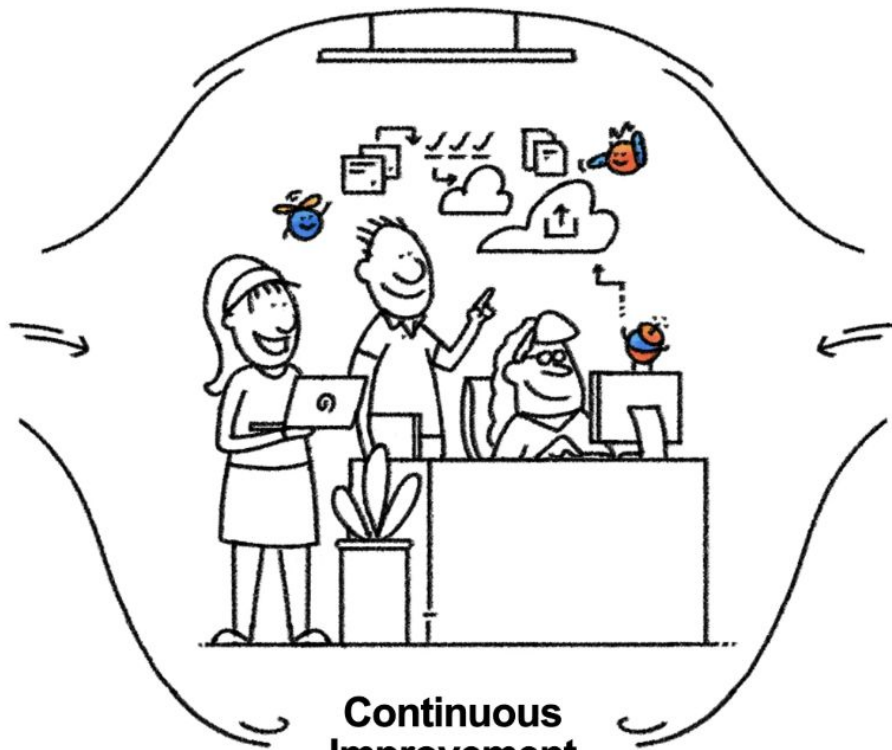
Integrated **Discovery** and **Automation**

The best thing for your processes



Process Owners

Optimize and monitor
process outcomes



Continuous Improvement

Increase efficiency and reduce
risk across the organization



Automation Leads

Scale and accelerate RPA
across the organization

Recap

- Discovery is the foundation for process improvement
- Multiple methods: interviews, workshops, observation, mining
- Importance of stakeholder identification and scoping
- Best practices: triangulation, validation, iteration
- Future is in automated continuous monitoring / learning